

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third/Fourth Semester of B. Tech. Examination (EE)

November 2013

EE-201 CONTROL ENGINEERING

Date: 11.11.2013, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q-1 (a) Define the following terms 02
- Control system, Open Loop control system
- (b) Explain closed loop control system by giving suitable example. Also enlist advantages for the same. 05
- Q-2 (a) Obtain the transfer function using Mason's Gain formulae for the signal flow graph shown in Figure I. 07

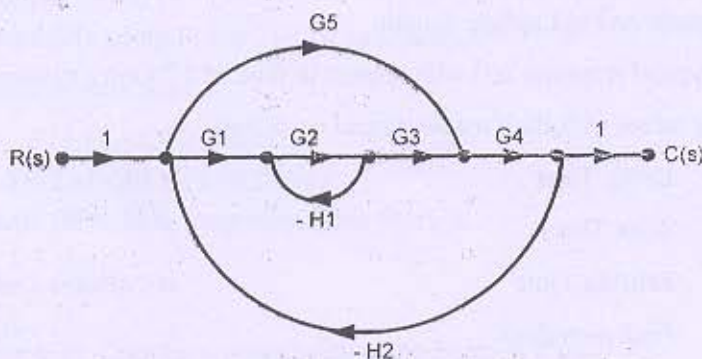


Figure – I

- (b) Explain Missile launching system with the help of neat figure. 07

OR

- Q-2 (a) (i) Define following terms. 04
- Forward path, Source node, Sink node, Chain node, Feedback loop
- (ii) Obtain T.F. for simple closed loop control system having forward path gain $G(s)$, Feedback path Gain $H(s)$ 03
- (b) Obtain Transfer Function using Block Diagram Reduction method for the block diagram shown in Figure II. 07

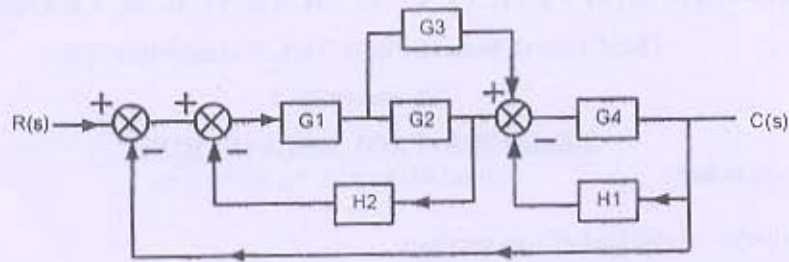


Figure - II

Q-3 (a) Explain PD controller in detail.

07

(b) For a unity feedback system has

07

$$G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$$

Determine (i) Type & Order of the system (ii) Error constants

(iii) Error for ramp input with magnitude 4.

OR

Q-3 (a) Explain in detail about Step, Ramp, Impulse functions. Also tabulate their values in Time domain and in Laplace domain.

07

(b) Draw a typical response $c(t)$ with respect to time of 2nd Order system and Define the following terms with their mathematical equations..

07

- Delay Time
- Rise Time
- Settling Time
- Peak overshoot

SECTION - II

Q-4 Obtain the transfer function from the given circuit shown in figure III.

07

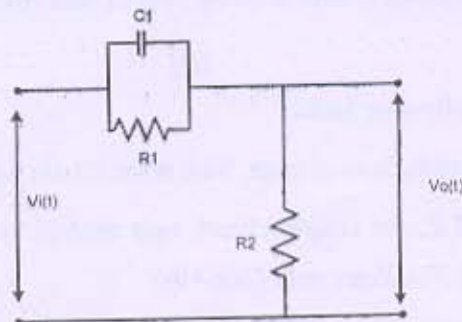


Figure III

- Q-5 (a) Draw the approximate root locus diagram for a closed loop control system whose open loop transfer function is given by 07

$$G(s)H(s) = \frac{K}{s(s+5)(s+2)}$$

Comment on stability.

- (b) The characteristics equation of the system is given by $s^6 + 4s^5 + 3s^4 - 16s^3 - 64s - 48 = 0$ 07
Find the number of roots of these equations with positive real part, zero real part and negative real part.

OR

- Q-5 (a) Sketch the root locus for a system having 07

$$G(s)H(s) = \frac{K}{s(s^2+2s+2)}$$

Also comment on stability.

- (b) The characteristics equation of the system is given by. 07

I. $s^3 + 6s^2 + 11s + 6 = 0$

II. $s^3 + 4s^2 + s + 16 = 0$

Comment on stability using Routh - Hurwitz criterion.

- Q-6 (a) A unity feedback system has 07

$$G(s) = \frac{K(1+0.2s)(1+0.025s)}{s^2(1+0.001s)(1+0.005s)}$$

Draw asymptotic Bode Plot. Determine Gain Margin.

- (b) A unity feedback system has 07

$$G(s) = \frac{10}{s(s+1)(s+2)}$$

Comment on stability using Nyquist Plot.

OR

- Q-6 (a) The Open Loop transfer function of unity Feed Back Control System is given by 07

$$G(s) = \frac{K}{s(1+0.001s)(1+0.25s)(1+0.1s)}$$

Determine the value of K so that the system will have a phase margin of 40°

What will be the gain margin? Use Bode Plot.

- (b) Explain Phase Lead Compensation technique in detail. 07
